

Ques No:36, Input=0, 15->11->2->3(repeat), all undesired state go to 11

IN	A	B	C	D	A+	B+	C+	D+	TA	DB	DC	JD	KD
0	0	0	0	0	1	0	1	1	1	0	1	1	X
0	0	0	0	1	1	0	1	1	1	0	1	X	0
0	0	0	1	0	0	0	1	1	0	0	1	1	X
0	0	0	1	1	1	1	1	1	1	1	1	X	0
0	0	1	0	0	1	0	1	1	1	0	1	1	X
0	0	1	0	1	1	0	1	1	1	0	1	X	0
0	0	1	1	0	1	0	1	1	1	0	1	1	X
0	0	1	1	1	1	0	1	1	1	0	1	X	0
0	1	0	0	0	1	0	1	1	0	0	1	1	X
0	1	0	0	1	1	0	1	1	0	0	1	X	0
0	1	0	1	0	1	0	1	1	0	0	1	1	X
0	1	0	1	1	0	0	1	0	1	0	1	X	1
0	1	1	0	0	1	0	1	1	0	0	1	1	X
0	1	1	0	1	1	0	1	1	0	0	1	X	0
0	1	1	1	0	1	0	1	1	0	0	1	1	X
0	1	1	1	1	1	0	1	1	0	0	1	X	0

Ques No:36, Input=1, 15->1->7->4(repeat), all undesired state go to 1

IN	A	B	C	D	A+	B+	C+	D+	TA	DB	DC	JD	KD
1	0	0	0	0	0	0	0	1	0	0	0	1	X
1	0	0	0	1	0	1	1	1	0	1	1	X	0
1	0	0	1	0	0	0	0	1	0	0	0	1	X
1	0	0	1	1	0	0	0	1	0	0	0	X	0
1	0	1	0	0	1	1	1	1	1	1	1	1	X
1	0	1	0	1	0	0	0	1	0	0	0	X	0
1	0	1	1	0	0	0	0	1	0	0	0	1	X
1	0	1	1	1	0	1	0	0	0	1	0	X	1
1	1	0	0	0	0	0	0	1	1	0	0	1	X
1	1	0	0	1	0	0	0	1	1	0	0	X	0
1	1	0	1	0	0	0	0	1	1	0	0	1	X
1	1	0	1	1	0	0	0	1	1	0	0	X	0
1	1	1	0	0	0	0	0	1	1	0	0	1	X
1	1	1	0	1	0	0	0	1	1	0	0	X	0
1	1	1	1	0	0	0	0	1	1	0	0	1	X
1	1	1	1	1	0	0	0	1	1	0	0	X	0

Flip – flop excitation table

T flip – flop excitation table

Q	Q+1	T
0	0	0
0	1	1
1	0	1
1	1	0

D flip – flop excitation table

Q	Q+1	D
0	0	0
0	1	1
1	0	0
1	1	1

JK flip – flop excitation table

Q	Q+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

K-map

INPUT=0

AB\CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	0	0	0	0
10	0	0	1	0

INPUT=1

AB\CD	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	1	1
10	1	1	1	1

$$\begin{aligned}
 TA &= \text{IN}'A'C' + \text{IN}'A'D + \text{IN}'A'B + \text{IN}'B'CD + \text{INA} + \text{INBC'D}' \\
 &= \text{IN}'(A'C' + A'D + A'B + B'CD) + \text{IN}(A + BC'D')
 \end{aligned}$$

DB

IN = 0

Group 1: $IN'A'B'CD$

AB/CD	00	01	11	10
00	0	0	1	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

IN = 1

Group 2: $INA'B'C'D$

AB/CD	00	01	11	10
00	0	1	0	0
01	1	0	1	0
11	0	0	0	0
10	0	0	0	0

Group 3: $INA'BC'D'$ Group 4: $INA'BCD$

$$= IN'A'B'CD + INA'B'C'D + INA'BC'D' + INA'BCD$$

$$= IN'A'B'CD + INA'D(B'C' + BC) + INA'BC'D'$$

$$= IN'A'B'CD + INA'D(B \oplus C)' + INA'BC'D'$$

DC

IN = 0

AB/CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

If $IN = 0$, output = 1
So $= IN'$

IN = 1

Group 1: $INA'B'C'D$

AB/CD	00	01	11	10
00	0	1	0	0
01	1	0	0	0
11	0	0	0	0
10	0	0	0	0

Group 2: $INA'BC'D'$

$$= IN' + INA'B'C'D + INA'BC'D'$$

$$= IN' + INA'C'(B'D + BD')$$

$$= IN' + INA'C'(B \oplus D)$$

JD

IN = 0

AB/CD	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

IN = 1

AB/CD	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

JD = 1, no matter IN=0 or IN=1

JD = 1

KD

IN = 0

AB/CD	00	01	11	10
00	X	0	0	X
01	X	0	0	X
11	X	0	0	X
10	X	0	1	X

Group 1: IN'AB'C

IN = 1

AB/CD	00	01	11	10
00	X	0	0	X
01	X	0	1	X
11	X	0	0	X
10	X	0	0	X

Group 2: INA'BC

= IN'AB'C + INA'BC